



Full Length Article

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Reduced Enterohepatic Recirculation of Mycophenolate and Lower Blood Concentrations Are Associated with the Stool Bacterial Microbiome after Hematopoietic Cell Transplantation



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A B S T R A C T

Mycophenolate mofetil (MMF) is an important immunosuppressant used after allogeneic hematopoietic cell transplantation (HCT). MMF has a narrow therapeutic index, and blood concentrations of mycophenolic acid (MPA), the active component of MMF, are highly variable. Low MPA concentrations are associated with the risk of graft-versus-host disease (GVHD), whereas high concentrations are associated with toxicity. Reasons for variability are not well known and may include the presence of β -glucuronidase-producing bacteria in the gastrointestinal tract, which enhance enterohepatic recirculation (EHR) by transforming MPA metabolites formed in the liver back to MPA. This study was conducted to determine whether individuals with high MPA EHR have a greater abundance of β -glucuronidase-producing bacteria in their stool and higher MPA concentrations compared with those with low EHR. We conducted a pharmacomicrobiomics study in 20 adult HCT recipients receiving a myeloablative or reduced-intensity preparative regimen. Participants received MMF 1 g i.v. every 8 hours with tacrolimus. Intensive pharmacokinetic sampling of MMF was conducted before hospital discharge; total MPA, MPA glucuronide (MPAG), and acyl-glucuronide metabolite (acylMPAG) were measured. EHR was defined as the ratio of MPA area under the concentration-versus-time curve (AUC)₄₋₈ to MPA AUC₀₋₈. Differences in stool microbiome diversity and composition, determined by shotgun metagenomic sequencing, were compared above and below the median EHR (22%; range, 5% to 44%). The median EHR was 12% in the low EHR group and 29% in the high EHR group. MPA troughs, MPA AUC₄₋₈, and acyl-glucuronide metabolite (acylMPAG) AUC₄₋₈/AUC₀₋₈ ratio were greater in the high EHR group compared with the low EHR group (1.53 μ g/mL versus .28 μ g/mL [$P = .0001$], 7.33 hour- μ g/mL versus 1.79 hour- μ g/mL [$P = .0003$], and .33 hour- μ g/mL versus .24 hour- μ g/mL [$P = .0007$], respectively). MPA AUC₀₋₈ was greater in the high EHR group than in the low EHR group, and the difference trended toward significance (22.8 hour- μ g/mL versus 15.3 hour- μ g/mL; $P = .06$). *Bacteroides vulgatus*, *Bacteroides stercoris*, and *Bacteroides thetaiotaomicron* were 1.2- to 2.4-fold more abundant ($P = .039$, .024, and .046, respectively) in the high EHR group. MPA EHR was positively correlated with *B. vulgatus* ($\rho = .58$; $P \leq .01$) and *B. thetaiotaomicron* ($\rho = .46$; $P < .05$) and negatively correlated with *Blautia hydrogenotrophica* ($\rho = -.53$; $P < .05$). Therapeutic MPA troughs were achieved in 80% of patients in the high EHR group but in no patients in the low EHR group. There was a trend toward differences in MPA AUC₀₋₈ and MPA concentration at steady-state (μ g/mL) between the high EHR group versus the low EHR group ($P = .06$). MPA EHR was variable. Patients with high MPA EHR had greater abundance of *Bacteroides* species in stool and higher MPA exposure compared with patients with low MPA EHR. Therefore, *Bacteroides* may be protective against poor outcomes, such as graft-versus-host disease, in some patients but may increase the risk of MPA adverse effects in others. These data need to be confirmed and studied after oral MMF therapy.

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INTRODUCTION

Numerous diseases have been linked to the gut microbiota, mainly through interactions with the host immune system. In hematopoietic cell transplantation (HCT), there is an

association between the stool microbiome—especially bacterial diversity—and risk of graft-versus-host disease (GVHD) [1–3]. Holler et al. [4] found that an increased abundance of *Enterococcus* species was associated with GVHD. Jenq et al. [5] showed that a highly diverse intestinal microbiota was associated with a lower incidence of GVHD-related mortality. In particular, the presence of the genus *Blautia* was associated with reduced mortality due to GVHD. Furthermore, GVHD was found to be significantly associated with decreasing nonpathogenic *Clostridia* abundance, possibly owing to clindamycin exposure [6].

It is well established that certain diseases and drugs profoundly alter the composition of the gut microbiome, leading to dysbiosis. HCT recipients are given antibiotics and nonantibiotic medications during the course of transplantation that can cause dysbiosis, which may adversely influence the effectiveness of other drugs and specifically mycophenolate [7,8]. Mycophenolate mofetil (MMF) is a common immunosuppressant that is critical for donor engraftment and prevention of acute GVHD (aGVHD). It reversibly inhibits inosine monophosphate dehydrogenase, which reduces T and B lymphocyte proliferation [9]. Despite the use of MMF and other potent immunosuppressants, aGVHD still occurs in up to 50% of patients and is associated with nonrelapse mortality [10]. Variability in immunosuppressant exposure is an important contributor to the failure of immunosuppressants, leading to underexposure and loss of immune control or overexposure and toxicity necessitating dosage reduction or discontinuation. Pharmacokinetic variability of mycophenolic acid (MPA), the active component of MMF, is high in HCT [11]. We and others have shown that low MPA concentrations are associated with poorer engraftment and higher incidence of aGVHD, whereas high concentrations are associated with a risk of toxicity [12–16].

The disposition and metabolism of MMF are complex. MPA undergoes enterohepatic recirculation (EHR) that is mediated by intestinal β -glucuronidases produced primarily by gram-negative anaerobes [17]. EHR is important because it contributes to maintaining MPA blood concentrations. Interpatient and inpatient variability in EHR may lead to unpredictable immunosuppression, which may adversely affect engraftment, GVHD protection, and MMF toxicity. Ex vivo culturing systems of microbial communities have identified microbiome-derived metabolism in 28 pharmacologic classes including MMF, resulting in parent drug depletion or the appearance of new metabolites [18]. In an in vitro microbiome model, MMF was one of the most highly metabolized drugs, with approximately 70 bacteria showing metabolic activity [19].

Zimmerman-Kogadeeva et al. [20] simulated host-microbiome deglucuronidation and glucuronyltransferase enzyme activity using experimental data and tested the conditions under which EHR affects drug exposure. They were able to predict the concentrations of drugs using these models, establishing the potential role of microbiome in the pharmacokinetics of MMF [20].

There are little data on EHR in HCT recipients. Disease and xenobiotic compounds used in patient populations change the composition and/or abundance of β -glucuronidase-forming bacteria in the gut microbiota. β -glucuronidase enzymes are produced primarily by gut microbes from Bacteroidetes (produces the largest number of enzymes), Firmicutes, Verrucomicrobia, and Proteobacteria phyla [21]. Any disruption in these organisms may reduce the amount of β -glucuronidase in the gut, reduce EHR, lower systemic MPA exposure, and increase the risk of MMF failure. In addition, MMF is associated with diarrhea, and in kidney transplant recipients, higher fecal β -glucuronidase activity is associated with a longer course of

diarrhea compared with lower β -glucuronidase activity [22]. No studies have investigated the effect of gut microbiota on MPA exposure in HCT recipients.

The primary objective of this study was to evaluate the association between the stool microbiome composition and abundance with MPA EHR and the pharmacokinetics of MPA and its metabolites. As a secondary objective, the association between microbiome β -glucuronidase gene abundance in the stool with MPA EHR was determined.

METHODS

Study Design and Participant Enrollment

In this prospective, observational pharmacomicrobiomics study, eligible participants were adults age 18 to 75 years with hematologic malignancies who underwent HLA-matched, haploidentical, or unrelated donor HCT after a reduced-intensity or myeloablative preparative regimen. All participants received post-transplantation cyclophosphamide for GVHD prophylaxis on post-transplantation days +3 and +4. Immunosuppression with MMF and tacrolimus was started on day +5. Participants were excluded who had undergone previous HCT or organ transplantation, had liver dysfunction (defined as total bilirubin concentration 2-fold above the upper limit of normal range), or had kidney dysfunction (defined as creatinine clearance [CrCl] <30 mL/minute). Standard antibiotic prophylaxis consisted of levofloxacin beginning on day –1 and continuing until neutrophil recovery. Cefepime was the recommended initial empiric antibiotic for neutropenic fever. All antibiotic use and changes were at the discretion of the treating physician. No patients received probiotics. The study protocol was approved by the University of Minnesota's Institutional Review Board (Study 00005621), and the study procedures adhered to the tenets of the Declaration of Helsinki. All participants provided informed consent for the use of their data.

Pharmacokinetic Sampling of MMF

Participants received MMF 1 g i.v. every 8 hours over 2 hours and tacrolimus .03 mg/kg i.v. over 24 hours beginning on day +5. Pharmacokinetic sampling was conducted on day +7 in the inpatient unit over one i.v. MMF dosing interval at steady state (a minimum of 3 unchanged doses). At the time of pharmacokinetic sampling, a CrCl >30 mL/minute and total bilirubin <3 times the upper limit of normal were required. If a patient's CrCl in renal or hepatic function was not within these parameters, the patient was excluded from the study and replaced. No patients were excluded because of these criteria. Whole blood samples (4 mL in a purple top tube) for pharmacokinetics analysis were obtained at 0, 2, 3, 4, 5, 6, 7, and 8 hours following the start of the 2-hour infusion.

Processing and Bioanalysis of MMF

MMF blood samples were transferred on ice in an insulated container to the laboratory within 1 hour of collection and processed. The samples were centrifuged at 2200 rpm for 10 minutes at 4 °C. After centrifugation, plasma (~5 to 1 mL) was transferred into a screw top vial containing 85% H₃PO₄ at a 1:50 ratio of 85% H₃PO₄ to plasma for total MPA, MPAG, and acylMPAG. Acidification was required to prevent the breakdown of the acylMPAG metabolite. Tubes were stored at –80 °C until being assayed for total MPA, MPAG, and acylMPAG by HPLC-MS [23,24]. The lower limits of quantification were 25 ng/mL for total MPA, 1 μ g/mL for MPAG, and 25 ng/mL for acylMPAG.

Pharmacokinetic Analysis and Determination of MPA EHR

MPA, MPAG, and acylMPAG concentration-time data were analyzed by noncompartmental analysis. WinNonLin Phoenix (Certara, Princeton, NJ) was used to calculate MPA, MPAG, and acylMPAG area under the curve (AUC) using the linear trapezoidal rule. Concentration at steady-state (C_{ss}) was determined by dividing AUC_{0–8} by the dosing interval in hours. The concentration-time curves of MPA were visually inspected in all patients. A second MPA peak, when present, was observed at ≥ 4 hours after the dose. Therefore, MPA EHR was defined as the ratio of MPA AUC_{4–8} to MPA AUC_{0–8} and was computed for each participant. Participants with MPA EHR above the median were considered to have high EHR, and those below the median were considered to have low EHR.

Stool Collection and Microbiome Analysis

Stool samples for DNA analysis were collected on the day of MMF pharmacokinetic sampling (± 48 hours) and stored at –80 °C after collection. Stool samples for microbiome characterization were collected by swabbing the stool samples obtained from fresh collection. The stool swabs were immediately placed in tubes containing 95% ethanol. The characterization of the microbiome and β -glucuronidase gene abundance were determined in each sample.

DNA was extracted from the swabbed stool samples using the DNeasy PowerSoil DNA Isolation Kit (QIAGEN, Hilden, Germany). The DNA was then

used for shotgun sequencing using 1/4 NexteraXT libraries [25] on an Illumina NextSeq instrument (Illumina, San Diego, CA). A mean read depth of 2.9 million raw reads (range, .4 to 5.7 million) per sample was achieved. Bioinformatic processing and annotations were done using MetaPhlAn3 and HUMAnN3 [26]. Functional genes were annotated against the MetaCyc [27] and Uniref [28] databases and classified using level 4 Enzyme Commission categories. Taxonomic data were reported as percent of total reads, and functional gene data were normalized as copies per million. Sequencing data have been deposited in the SRA archive under BioProject accession no. SRP303698.

Statistical Analysis

The Wilcoxon rank-sum test was used to test the differences in pharmacokinetic measures by EHR group at a .05 significance level. Species composition data were reported as relative abundances, and abundances of β -glucuronidase genes were normalized as copies per million. Differences among microbial communities were determined based on Bray-Curtis dissimilarities calculated from species abundance tables and evaluated by analysis of similarity using vegan [29]. Communities were visualized by ordination using principal coordinate analysis, and species that were significantly correlated with axis position (Spearman correlation) were overlaid using the corr.axes command in mothur [30]. Differences in relative abundances of species, β -glucuronidase gene abundance, and alpha diversity measured by the Shannon index were determined using the Kruskal-Wallis test. All statistics were evaluated at $\alpha = .05$.

RESULTS

Twenty HCT recipients (12 males and 8 females) were studied; their demographic characteristics are presented in Table 1. The median serum creatinine and total bilirubin concentrations on the day of pharmacokinetic sampling were .58 (interquartile range [IQR], .16) mg/dL and .55 (IQR, .32) mg/dL, respectively. Pharmacokinetic sampling was performed at a median of day 7 (range, day 6 to 8) post-transplantation. The median plasma MPA, MPAG, and acylMPAG AUC₀₋₈ values in the total cohort were 19.36 hours· μ g/mL (IQR, 1.5 hours· μ g/mL), 207.69 hours· μ g/mL (IQR, 109.95 hours· μ g/mL), and 2.06 hours· μ g/mL (IQR, 1.52 hours· μ g/mL), respectively.

Table 1
Patient Demographics (N = 20)

Characteristic	Value
Age, yr, median (IQR)	46 (19.5)
Male sex, n (%)	12 (60)
Race, n (%)	
American Indian	2 (10)
Caucasian	18 (90)
Diagnosis, n (%)	
Acute leukemia	16 (80)
Lymphoma	2 (10)
Myelodysplastic syndrome	2 (10)
Preparative regimen, n (%)	
Reduced-intensity	5 (25)
Myeloablative	15 (75)
Height on admission day, m, median (IQR)	1.75 (.14)
Weight on admission day, kg, median (IQR)	90.7 (27.1)
Laboratory values at the time of pharmacokinetic sampling, median (IQR)	
Serum creatinine, mg/dL	.58 (.16)
Total bilirubin, mg/dL	.55 (.32)
Blood urea nitrogen, mg/dL	9.5 (4.5)
Aspartate aminotransferase, U/L	10.5 (4.75)
Alanine transaminase, U/L	21 (17)
Alkaline phosphatase, U/L	67 (19.75)

Pharmacokinetics by EHR Classification

MPA, MPAG, and acylMPAG AUC₀₋₄, AUC₄₋₈, and AUC₀₋₈ were calculated (Figure 1 and Supplementary Table S1). The EHR was the ratio of MPA AUC₄₋₈/MPA AUC₀₋₈, and the overall median was .22 (range, .05 to .44). The median ratio was .29 in the high EHR group and .12 in the low EHR group (Figure 1, Supplementary Table 1). The MPA trough at 8 hours post dose was greater in the high EHR compared with the low EHR group (1.53 μ g/mL versus .28 μ g/mL; $P = .0001$). The MPA AUC₄₋₈ was higher in the high EHR group (7.33 hours· μ g/mL versus 1.79 hours· μ g/mL; $P = .0003$). The median acylMPAG AUC₄₋₈/AUC₀₋₈ ratio was higher in those classified as high EHR compared with those classified as low EHR (.33 hour· μ g/mL versus .24 hour· μ g/mL; $P = .0007$). There was a trend toward differences in both MPA AUC₀₋₈ and MPA C_{ss} μ g/mL in the high EHR group versus the low EHR group ($P = .06$). There were differences in the MPA concentration-time profiles, with higher concentrations in the high EHR group beginning at around 4 hours (Figure 2). In the high EHR group, 80% of the participants had MPA trough concentrations within the therapeutic range, compared with none in the low EHR group (Table 2).

Microbiome Association with EHR

The community composition of stool samples varied significantly between patients with high EHR and those with low EHR (analysis of similarity, $R = .15$; $P = .012$) (Figure 3A-C). *Bacteroides vulgatus*, *Bacteroides stercoris*, and *Bacteroides thetaiotaomicron* had significantly greater relative abundances in the high EHR group ($P = .039$, .024, and .046, respectively) (Figure 3D-F). MPA EHR was also positively correlated with the presence of *B. vulgatus* ($\rho = .58$; $P = .01$) and *B. thetaiotaomicron* ($\rho = .46$; $P = .04$) (Table 3). *Blautia hydrogenotrophica* abundance was inversely correlated with MPA EHR ($\rho = -.53$; $P = .02$), MPAG AUC₄₋₈/AUC₀₋₈ ratio ($\rho = -.57$; $P = .01$), acylMPAG AUC₄₋₈/AUC₀₋₈ ratio ($\rho = -.50$; $P = .03$), and MPA C_{ss} ($\rho = -.50$; $P = .03$). *Parabacteroides distasonis* abundance was significantly and positively correlated with MPAG AUC₄₋₈/AUC₀₋₈ ratio ($\rho = .69$; $P = .001$) and MPA C_{ss} ($\rho = .53$; $P = .02$). *Bacteroides uniformis* abundance was positively correlated with MPAG AUC₀₋₈ ($\rho = .48$; $P = .04$). The abundances of genes annotated as β -glucuronidase did not differ significantly based on patients' EHR status ($P = .496$, Kruskal-Wallis test), although a greater variability in gene abundances was observed in patients with low EHR (Figure 3G). There was no significant difference in alpha diversity (Shannon index) between the high EHR and low EHR groups (Figure 3H), which is surprising given the higher antibiotic burden in the low EHR group. This may be related to the use of a relatively small shotgun sequencing depth and the need for extensively deeper sequencing for comprehensive community characterization [31].

More individuals in the high EHR group than in the low EHR group received levofloxacin prophylaxis in the 72 hours before pharmacokinetic sampling (80% versus 30%) (Table 4). In the low EHR group, 8 of 10 participants (80%) received i.v. cefepime, cefpodoxime, cefdinir, azithromycin, and/or vancomycin before pharmacokinetic sampling, compared with only 2 of 10 (20%) in the high EHR group.

DISCUSSION

EHR is an important pharmacologic process in which biliary-excreted drugs are reabsorbed in the intestine instead of being eliminated. MMF is a classic example of a recirculated drug. Multiple gut bacteria produce β -glucuronidase, and changes in abundance or composition of these organisms may affect MMF recirculation and systemic exposure. In this study,

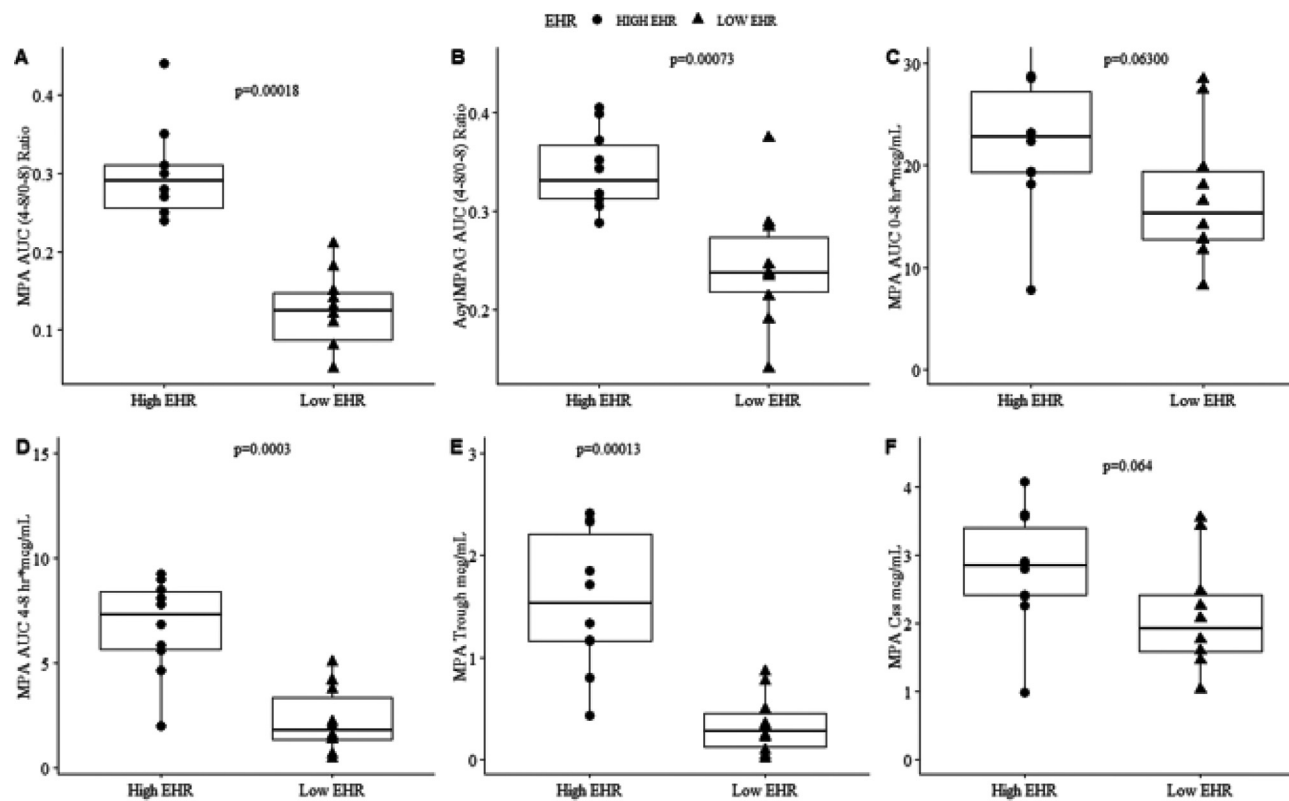


Figure 1. MMF pharmacokinetics by EHR group. Circles and triangles represent individual values of high EHR and low EHR, respectively.

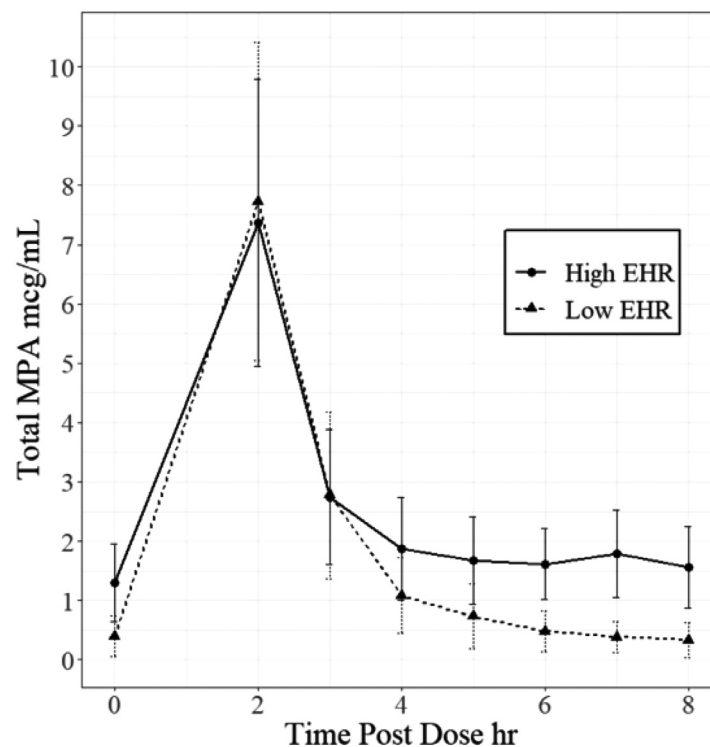


Figure 2. Total MPA concentration-time profile in patients with high and low EHR. Data are mean with SD. The median AUC_{0-8} was 22.82 hour· μ g/mL in the high EHR group and 15.34 hour· μ g/mL in the low EHR group ($P = .06$), and the median MPA AUC_{4-8} was 7.33 hour· μ g/mL and 1.79 hour· μ g/mL, respectively ($P = .0003$).

we demonstrated that β -glucuronidase producing bacteria and others may be associated with changes in MPA recirculation. When given orally, MPA is absorbed primarily in the stomach and proximal small intestine [9], whereas when given i.v., it bypasses first-pass metabolism. MPA is biotransformed in the liver primarily by uridine 5'-diphospho-glucuronosyltransferases (UGTs) 1A9 and 1A8 to the inactive metabolite MPAG and to a lesser extent the active metabolite acylMPAG by UGT2B7 [17]. MPAG and possibly acylMPAG are excreted renally and into the bile, where they are subject to EHR. From the bile, MPAG is excreted into the small intestine, where deconjugation occurs via β -glucuronidases and active MPA is reformed. After deconjugation, MPA is reabsorbed back into the circulation from the distal regions of the small intestine [17,32]. In normal volunteers, EHR is a substantial portion of overall exposure and is estimated to be 37% of the MPA AUC (range, 10% to 61%) [33].

Therapeutic concentrations of MPA have been shown to be important for engraftment and aGVHD and were consistently lower in the low EHR group. No patients with low MPA EHR had a therapeutic MPA trough ($>1 \mu$ g/mL), whereas 80% of those in the high EHR group had a therapeutic MPA trough. The median MPA AUC_{0-8} was lower in the low EHR group compared with the high EHR group (15.35 hour· μ g/mL versus

22.82 hour· μ g/mL; $P = .06$). The MPA C_{ss} was 2.85 μ g/mL in the high EHR group and 1.92 μ g/mL in the low EHR group ($P = .06$). In our previous studies, we found that MPA trough, AUC, and C_{ss} targets are difficult to attain even at doses of 3 g/day, and altered EHR likely contributes to this difficulty [34].

In the present study, we found that greater abundances of the species *B. vulgatus*, *B. stercoris*, and *B. thetaiotaomicron* were associated with higher MPA EHR. These anaerobic, gram-negative rods likely produce β -glucuronidase, which deconjugates MPAG in the intestine to MPA and is then readily enterohepatically recirculated [35]. In a HCT microbiome study, the *Bacteroidetes* phylum was more abundant in children without aGVHD compared with those who developed aGVHD [36]. This is consistent with the presence of more β -glucuronidase enzyme, higher MPA EHR, and greater GVHD protection. *Bacteroides* are important in humans and compose a large proportion of bacteria in the human colon. They produce short-chain fatty acids, which are used by the host as an important energy source and also have other functions, such as providing protection from pathogens and nutrients to other microbial residents. The 3 individuals with the lowest MPA EHR had no detectable *Bacteroides* (Figure 3A).

We also found that relative abundances of *P. distasonis* were positively correlated with the MPAG ratio and MPA average concentration over a dosing interval (C_{ss}). *P. distasonis* grows in bile, and in mice it alters the production of secondary bile acids and increases levels of lithocholic acid and ursodeoxycholic acid [37]. MPAG is excreted into the gallbladder from the liver, and efficient deconjugation of MPAG and formation of MPA may be dependent on sufficient primary bile acid production. Thus, the presence of *P. distasonis* may represent good hepatobiliary system functioning. *B. uniformis* is positively correlated with MPAG AUC_{0-8} . This organism and other β -glucuronidase-producing bacteria are associated with the

Table 2
MPA Troughs by EHR Group

Trough	High EHR (N = 10), n (%)	Low EHR (N = 10), n (%)
Trough $<1 \mu$ /mL (below)	2 (20)	10 (100)
Trough 1–3.5 μ /mL (within)	8 (80)	0 (0)
Trough $>3.5 \mu$ /mL (above)	0 (0)	0 (0)

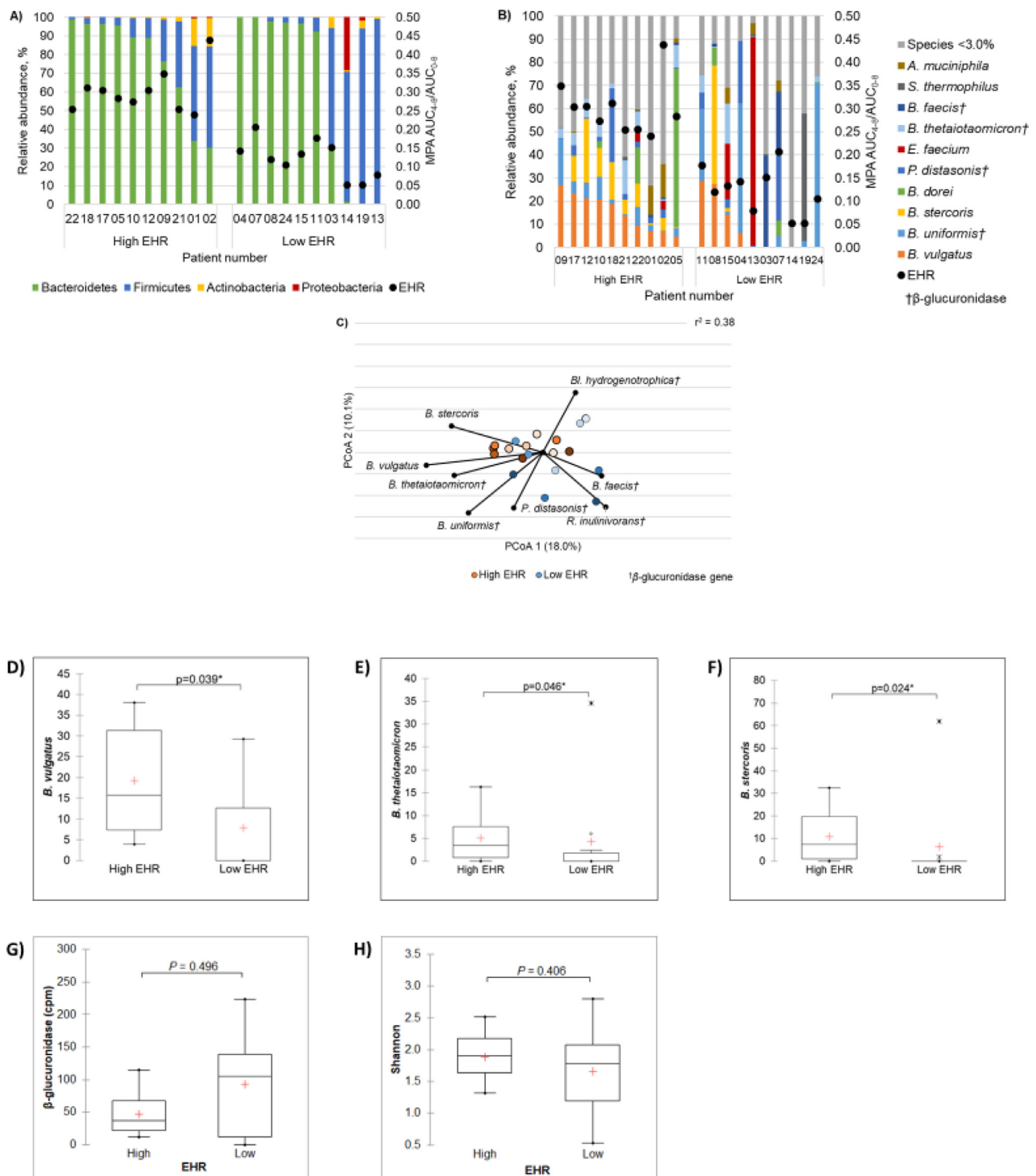


Figure 3. Microbial community composition and functional data. (A) Relative abundances of phyla. The x-axis represents patient ID number and black dots represent the EHR values. (B) Distribution of abundant species in individual patient samples. Species with a mean relative abundance <3.0% among all samples were consolidated for clarity, and species in which a β-glucuronidase gene was annotated are indicated in shades of blue. The x-axis represents patient number and black dots represent the EHR value in each patient. (C) Principal coordinate analysis of Bray-Curtis dissimilarities. Species shown reflect predominant species that were significantly correlated with axis position (Spearman correlation $P < .05$, where the distance from the origin indicates the strength of correlation). Orange shading indicates individuals with high EHR values; the darker the shading, the higher the EHR value. Blue shading indicates individuals with low EHR values; again, the darker the shading, the higher the EHR value. (D-F) Significant differences in *Bacteroides* spp. abundance by EHR group. (G) β-glucuronidase abundance. (H) Shannon index. Box-and-whisker plots show IQR ± standard error, where + indicates the mean. Statistics reflect the Kruskal-Wallis test.

deconjugation of SN-38 glucuronide, which during EHR is transformed back to SN-38, the active metabolite of irinotecan, resulting in greater toxicity [38].

We identified that *Bl. hydrogenotrophica* is inversely associated with 3 pharmacokinetic measures—MPA EHR, MPAG, and acylMPAG ratios—and MPA concentration over the dosing

Table 3

Significant Correlations of Species Relative Abundance with MMF Pharmacokinetics

Species	MPA AUC ₄₋₈ /AUC ₀₋₈	MPAG AUC ₄₋₈ /AUC ₀₋₈	AcyMPAG AUC ₄₋₈ /AUC ₀₋₈	MPAG AUC ₀₋₈	MPA C _{ss}
<i>B. vulgatus</i>	$\rho = .58^{**}$	NS	NS	NS	NS
<i>B. thetaiotaomicron</i>	$\rho = .46^*$	NS	NS	NS	NS
<i>Bl. hydrogenotrophica</i>	$\rho = -.53^*$	$\rho = -.57^*$	$\rho = -.50^*$	NS	$\rho = -.50^*$
<i>P. distasonis</i>	NS	$\rho = .69^{**}$	NS	NS	$\rho = .53^*$
<i>B. uniformis</i>	NS	NS	NS	$\rho = .48^*$	NS

Only species with at least 1 significant association are shown.

** $P = \text{or} < .01$; * $P < .05$; NS, not significant

interval (C_{ss}). A low abundance of *Blautia* spp. has been associated with aGVHD [39]. Our findings of lower MPA exposure in patients with *Blautia* spp. are inconsistent with greater protection from GVHD. Others have found that a loss of *Blautia* spp. was associated with the use of antibiotics active against anaerobic bacteria, and in our cohort, the low MPA EHR group received more antibiotics, including those active against anaerobes.

Not all *Bacteroides* spp. produce the β -glucuronidase enzyme. We found that the non- β -glucuronidase-producing *B. vulgatus* [40] was positively correlated with MPA EHR, although the mechanism of this association is unclear. We did not identify any associations with certain other β -glucuronidase-producing bacteria, such as *B. fragilis* (a major contributor to enzyme levels), *Escherichia coli*, *Parabacteroides merdae*, *Clostridium perfringens*, and *Enterococcus* spp. [21], although it does not exclude their effect because it may be obscured by more abundant species and observable in a larger study population.

In our cohort, 80% of the patients in the low EHR group received broad-spectrum i.v. cephalosporins and/or vancomycin in the 72 hours before pharmacokinetic sampling, compared with only 20% in the high EHR group. Broad-spectrum cephalosporins and vancomycin are active against gram-negative anaerobes, which is consistent with a reduction in β -glucuronidase enzyme and subsequently lower EHR. Antibiotic selection may have a substantial influence on EHR in HCT, although the extent and duration of antibiotic exposure needed to alter MPA EHR are unknown. In a previous study by our group in patients with acute leukemia and HCT, there was no change in bacterial alpha diversity and no major effects on abundance except for a slightly lower abundance of *Parabacteroides* spp. and a higher abundance of *Blautia* spp. in the levofloxacin group compared with no antibiotics [41]. However, we found that a greater abundance of *Blautia* spp. is correlated with lower EHR, inconsistent with it being a more beneficial taxon. Naderer et al. [42] showed that MPA AUCs were reduced by 33% when coadministered with norfloxacin and metronidazole, which is highly active against gram-negative anaerobes. In kidney transplant recipients, the use of oral ciprofloxacin or

amoxicillin-clavulanic acid resulted in a 46% reduction in MPA trough level within 72 hours of antibiotic initiation, suggesting that only minimal antibiotic exposure alters MPA EHR and blood concentrations [43]. MPA concentrations increase over time and decrease by up to ~50% when MPA is administered with amoxicillin-clavulanate, metronidazole, and ciprofloxacin [44]. In HCT, after reduced-intensity conditioning, MPA exposure has been shown to increase by nearly 40% in the first week and trended toward higher exposure at 1 month post-transplantation [45]. These longitudinal changes may be related in part to antibiotic prophylaxis, changes in antibiotic treatment, and/or recovery of the microbiome post-transplantation. In a recent mouse study, MMF was seen to alter the gut microbiome by selecting for β -glucuronidase-producing bacteria [46]. Vancomycin eliminated these bacteria, reduced MMF-related gastrointestinal toxicity, and increased stool MPAG concentrations but reduced stool MPA concentrations.

Most evaluations of MMF EHR are with oral administration. In this study, we evaluated i.v. MMF and found a median EHR of 22% (range, 5% to 44%). Previous studies with oral MMF in normal volunteers showed a higher MPA EHR when using cholestyramine as a probe (mean, 37%; range, 10% to 61%) [33]. Numerous drugs undergo EHR, and in the case of morphine, there is significantly greater EHR with oral administration compared with i.v. administration [47]. We expect that EHR variability and the influence of the bacterial microbiome may be more profound with oral MMF [33]. In an older HCT study, no MPA EHR was observed after oral administration, as secondary peaks were not seen; however, all patients were receiving concomitant cyclosporine use, which is known to greatly diminish or eliminate MPA EHR [32,45].

Multiple bacteria may be involved in MPA EHR. In an in vitro microbiome model, MMF was one of the most highly metabolized drugs by bacteria, with ~70 gut bacteria showing metabolic activity, including *Clostridium* spp.; numerous *Bacteroides* species, including *B. vulgatus*, *B. thetaiotaomicron*, and *B. stercoris*; *E. coli*; *Bifidobacterium adolescentis*; *Blautia* spp.; and *Ruminococcus gnavus* [19]. We also found that *B. vulgatus*, *B. thetaiotaomicron*, and *B. stercoris* were associated with MMF and specifically with higher EHR. Javdan et al. [18] found in a metabolism screen that gut microbiota were associated with ester hydrolysis of MMF, forming MPA. Many different β -glucuronidase enzymes are produced by bacteria. Data from the Human Microbiome Project from volunteer samples show a median of 23 (range, 4 to 38) unique β -glucuronidase protein sequences per individual [21]. There may be even greater numbers in immunocompromised populations, who may receive antibiotics that select for a greater abundance of anaerobic populations. Some β -glucuronidases are substrate-specific, and thus it is possible that not all bacteria producing this enzyme will affect MPA disposition [21]. Our analysis does not discern the enzyme(s) responsible for MPAG deconjugation and EHR, although many likely are involved.

Table 4

Antibiotic Exposure at 72 Hours before MPA Pharmacokinetic Sampling by EHR Group

High EHR (no. of Patients)	Low EHR (no. of Patients)
Levofloxacin (8)	Levofloxacin (3)
Cefepime (2)	Cefepime (1)
	Vancomycin (1)
	Azithromycin/cefepime (1)
	Vancomycin/cefepime (3)
	Cefpodoxime (1)
	Cefdinir (1)

There are limitations to our study. We used the median MPA EHR (22%) as a convenient cutoff point to stratify patients into high EHR and low EHR groups. The lowest and highest acceptable bounds for optimal EHR should be explored in larger samples. Although clinical factors and drug-drug interactions known to impact MPA concentrations were not evaluated in this study, their clinical impact is generally considered low except for the interaction with cyclosporine, and all patients in our study received tacrolimus. Our data indicate that despite similar abundance of the β -glucuronidase gene in the stool of patients with high EHR and low EHR, only several *Bacteroides* spp. showed significant correlations and differences in abundance between these groups. This may mean that the gene is not expressed in all *Bacteroides* spp. or that only some β -glucuronidase enzymes produced by *Bacteroides* spp. are active toward MPA. MMF is reabsorbed in the small intestine, but our stool samples primarily reflect the bacteria in the lower gastrointestinal tract; however, obtaining samples from the small intestine is infeasible [48].

Here we describe an association between the stool microbiome and MPA EHR after HCT. MMF is given as a fixed dose (ie, 2 to 3 g/day) [49], and therapeutic drug monitoring is not routinely performed in HCT; therefore, low or high MPA exposure influenced by the microbiome may go undetected. High EHR may be predictive of better immunosuppression but also may be related to MPA concentration-dependent toxicities, such as leukopenia and anemia. Testing of the microbiome may provide a tool to help guide initial dose selection and dose adjustments of MMF.

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SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.jctc.2022.04.018.

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